

Module 06: Namespaces, classes, fields and properties

Objective: Creation of a class and instantiation of objects

System requirements: Net 5.0

Tools: Visual Studio 2019

Duration: ~30min

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1 Creation of a class

A class Vehicle is to be created, which contains the following members:

Properties/Fields

- Name (string)
- MaximumSpeed (int)
- CurrentSpeed (int)
- Price (double)
- MotorRunning (bool)

Methods

- Info() => Returns a nicely formatted string that gives info about the vehicle
- StartMotor() => Changes MotorRunning to true; motor must not be running beforehand
- StopMotor() => Changes MotorRunning to false; vehicle must be stopped & motor must run
- Accelerate(int a) => Changes the current speed by the given value; The current speed must not fall below 0 and must not exceed the maximum speed

Constructor

- The constructor shall receive the name, the maximum speed and the price as passing parameters. The remaining properties are to be set so that when a new vehicle is created, the engine is off and it does not move.

2 Instantiation of objects

Now instantiate a vehicle (or several) in the Main() method and play around with it a bit. Let the Info() method output, accelerate the vehicle and test if it behaves as it should.

```
VW costs 2499€ and has a maximum velocity of 210km/h.
The motor of VW has been started.
VW now moves at 150km/h.
VW costs 2499€ and is driving with a velocity of 150km/h with a maximum of 210km/h.
```

```
Console.OutputEncoding = System.Text.Encoding.UTF8;
```